

CHAPTER 21 Study Guide - Electrochemistry

Key Concepts

21.1 Electrochemical Cells

- For any two metals in an activity series, the more active metal is the more readily oxidized.
- All electrochemical processes involve redox reactions.
- Energy is produced in a voltaic cell by spontaneous redox reactions within the cell.
- Current electrochemical technologies used to produce electrical energy include dry cells, lead storage batteries, and fuel cells.

21.2 Half-Cells and Cell Potentials

- The electrical potential of a cell results from a competition for electrons between two half-cells.
- The standard reduction potential of the hydrogen electrode has been assigned a value of 0.00 V.
- You can determine the standard reduction potential of a half-cell by using a standard hydrogen electrode and the equation for standard cell potential.
- If the cell potential for a given redox reaction is positive, then the reaction is spontaneous. If the cell potential is negative, then the reaction is nonspontaneous.

21.3 Electrolytic Cells

- The key difference between voltaic and electrolytic cells is that in a voltaic cell, the flow of electrons is the result of a spontaneous redox reaction, whereas in an electrolytic cell, electrons are pushed by an outside power source, such as a battery.
- The products of the electrolysis of water are hydrogen gas and oxygen gas.
- During electrolysis of brine, chloride ions are oxidized to produce chlorine gas at the anode; water is reduced at the cathode to produce hydrogen gas.
- Electrolytic cells are commonly used in the plating, purifying, and refining of metals.

Vocabulary

- anode (p. 665)
- battery (p. 668)
- cathode (p. 665)
- cell potential (p. 671)
- dry cell (p. 667)
- electrical potential (p. 671)
- electrochemical cell (p. 665)
- electrochemical process (p. 665)
- electrode (p. 665)
- electrolysis (p. 678)
- electrolytic cell (p. 678)
- fuel cell (p. 669)
- half-cell (p. 665)
- reduction potential (p. 671)
- salt bridge (p. 665)
- standard cell potential (p. 672)
- standard hydrogen electrode (p. 672)
- voltaic cell (p. 665)

Key Equation

- $E^0_{\text{cell}} = E^0_{\text{REDUCTION}} - E^0_{\text{OXIDATION}}$